Team #8
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Section K3
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Transcribing Chess Board

We built an 8x8 chessboard that is able to electronically transcribe the piece locations and send it to a computer for further processing. The original goal was to be able to detect all the pieces and also have LEDs in each square to be able to highlight them (for opponent moves, possible moves, chess engine moves, etc.), but due to time constraints we skipped the LED part. We also realized very late that the keyboard matrix we designed has issues with phantom key presses and cannot detect accurately when multiple pieces are on the board at the same time. This could be fixed by having diodes in the matrix between the row and column channels but due to lack of time we couldn't implement the fix (Refer Figure 1). We faced unexpected issues when laser cutting the chess board: the laser didn't consistently cut through the wood, especially since we had a lot of small things to be cut on the board. We had to use multiple pieces of 12x24 inch^2 wood to compensate. Though the original design would've required only one. The final design can detect a few pieces fairly well if the piece applies enough pressure on the board, better and stronger magnets in the board could improve robustness. The laser cut pieces are borrowed from: https://www.dxfdownloads.com/free-2d-dxf-laser-cut-chess-set/

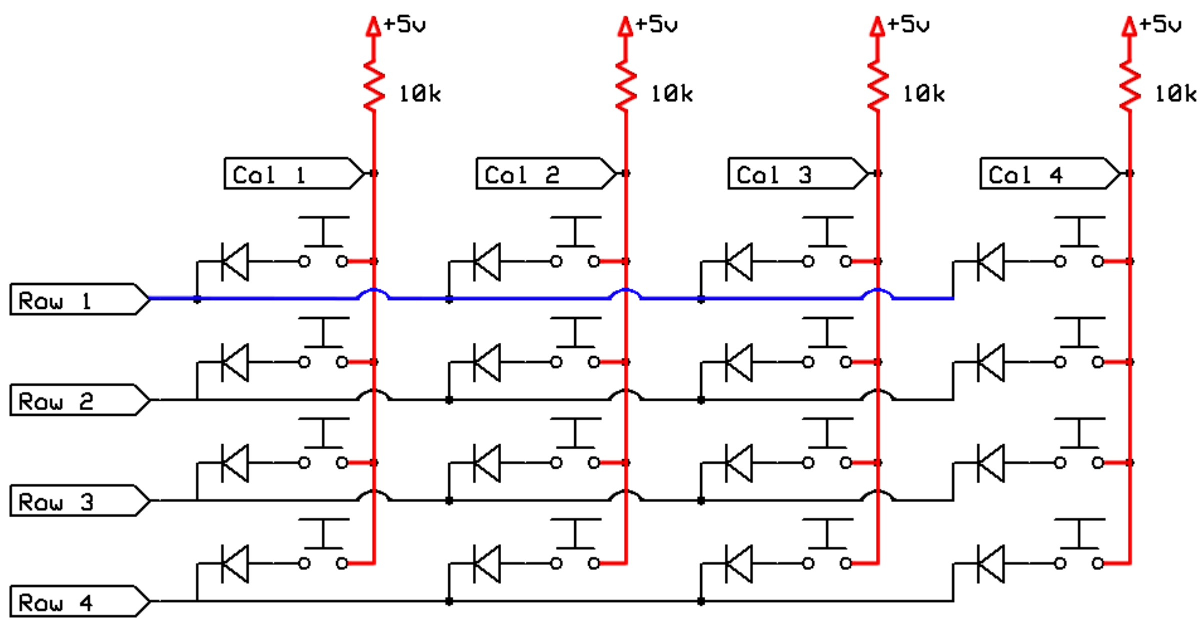

Figure 1: 4x4 Keyboard Matrix to detect multiple key presses at the same time
Source: https://www.gammon.com.au/forum/?id=14175

The detection code on Arduino works by setting a drive pin to LOW and reading off the sense pins. It then compares with the previous state and sends any changes to the PC (more detail in submitted .ino file).

```
…snipped initialization code…

void senseGrid() {
  for (int i = 0; i < N; i++) {
    pinMode(X_PINS[i], OUTPUT);
    digitalWrite(X_PINS[i], LOW);
    delay(SENSE_DELAY);
    for (int j = 0; j < N; j++) {
      GRID[i][j] = digitalRead(Y_PINS[j]);
    }
    pinMode(X_PINS[i], INPUT);
    delay(SENSE_DELAY);
  }
}

…other functions…

void setup() {
  … initialize pins and Serial …
}

void loop() {
  senseGrid(); // sense current state of grid
  compareGrid(); // compare with previous state and send any changes
  storeGrid(); // store current state for next iteration
  delay(DEBOUNCE_DELAY); // wait
}
```

Our team members performed the following tasks:

Siddharth Sahay	Laser Cut and assembled the chess pieces and wrote the code for the Arduino and the python program
Shashank Obla	Laser-Cut the board and added contacts and wiring to the board, parts ordering and team sheet updates